

Linear Algebra

Project



Due date: 16 July (2024)

In the digital age, consumers often compare various products online before making a purchase to ensure they get the best value for their money. This project will help you apply linear algebra concepts to real-world data by extracting features of products from e-commerce websites and using these features to predict prices with linear regression. This hands-on project will enhance your understanding of data analysis, regression techniques, and the practical applications of linear algebra.

Your tasks:

Step 1: Data Collection and Preparation

Selection of Product Category: Choose a specific category of products, such as electronics, books, or clothing.

Web Scraping: Use web scraping tools (e.g., BeautifulSoup, Scrapy, Selenium) to gather data from websites like Digikala, Divar or Bama. Ensure compliance with the website's terms of service.

Feature Extraction: Extract key features such as product name, price, rating, number of reviews, and other relevant attributes that may influence the price.

Data Transformation: Convert categorical data into numerical format using techniques like one-hot encoding.

Step 2: Linear Regression Model

Data Splitting: Split the data into training and testing sets.

Model Training: Fit a linear regression model using the training set.

Model Evaluation: Evaluate the model's performance on the testing set using metrics like Mean Squared Error (MSE).

Step 3: Price Prediction

Prediction: Use the trained model to predict prices for a new set of product features.

Notes:

Allowed programming languages: Python

Your implementation reports should be in a PDF file including an explanation of your approach, key points of your implementation, and report of your final results.

You should upload your submissions at Quera. All of the files should be in a ZIP file named in this format: “`LastName_SudentNumber.zip`”. Example: “`Zamani_40230400.zip`”

Any sign of cheating would result in a zero grade for this assignment.
